

Multi-Sensor Smoke & Fire Alarm with Integrated People Sensing & Counting

Pufferfish Security

MARK TIBBS (29191802), VITALIJA MARTINKUTE (29030002), DAWID KOWALSKI (29816244),
MOLLIE NAIMH ALLEN (29868067) & KAROL CABROWSKI (29966343)

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Executive Summary

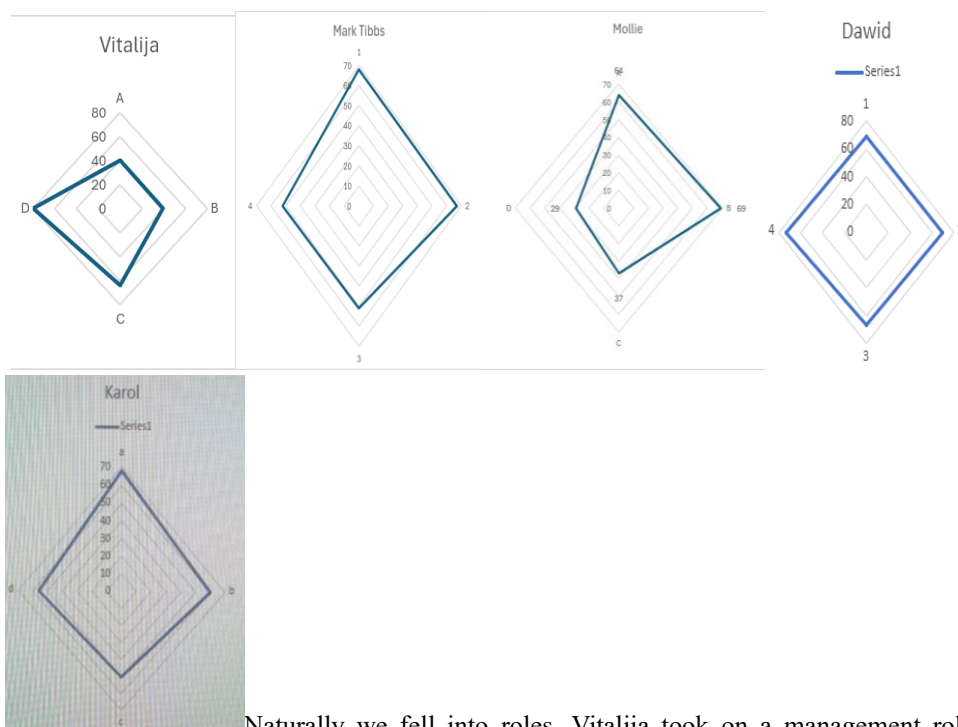
At Pufferfish Security, the challenge that we are addressing is the issue that inside commercial situations, companies use multiple different sensors, cameras, and safety systems regarding fire safety instead of using one integrated system. Therefore, our system includes a diverse and well-integrated range of sensors which relate to fire safety and prevention. To achieve this we first decided roles and spoke about what was the most fitting role for each member, discussing strengths and weaknesses as well as what each member enjoyed, we also decided very early on what to do as the project, leaving us ample time to produce each document and plan without fear for the timescale. We decided on the fire alarm with multiple sensors, then set out what would be the basic requirements like the people counter and motion detection systems to make our system unique to the market. We then set and completed tasks such as the Gantt chart, risk assessment, and others, before discussing ideas for the video. After this we finally developed and refined our final concept, selecting the most suitable components, materials and technology to create a realistic fire alarm system.

Team Dynamics & Roles

This project was our very first time coming together as a team, therefore we needed to set some groundwork. At the very beginning of the project, each member of our team took place in a quiz to verify their skills and where they would be the most comfortable within the team, of which the results are in the appendix. Using this quiz we correctly and affectively allocated jobs to each of our members.

According to Blackboard:

- A-preference people are analytical, rational, technical, logical, factual, and quantitative.
- B-preference people are procedural, scheduled, conservative, organised, sequential, reliable, tactical, and administrative.
- C-preference people are supportive, interpersonal, expressive, sensitive, symbolic, musical, and reaching out.
- D-preference people are more strategic, visual, imaginative, conceptual, and simultaneous.



Naturally we fell into roles, Vitalija took on a management role allocating tasks and recognising what needed to be done and when, Dawid took on the role of CAD modelling and design of 3D objects, Mark took on the role of organising and compiling information and documents and editing the final video, Mollie took on the finer details such as material selections, safeguarding/legislations and cost analysis, and Karol took on the role of generalised research across multiple tasks and feasibility analysis of product cost and technology.

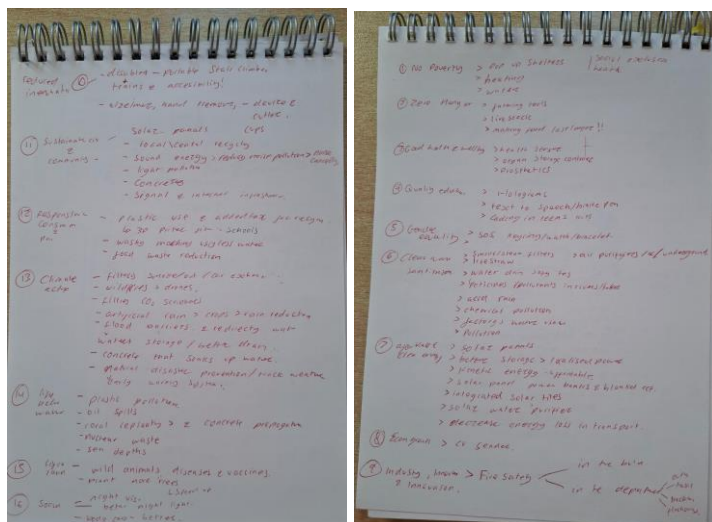
In our project, we used different project management activities which included planning, organisation, communication and monitoring progress. Our group consisted of five people where we met up every week and

discussed ideas on how to best create this fire alarm. During each of these sessions, we collectively discussed what needed to be completed and set deadlines that we all needed to meet by the following week. This allowed us to keep everything on schedule, so we ensure that everything was done before the deadline date. We kept track of every meeting with meeting notes which can be seen in the *appendix*.

Secondly, we have also formed a WhatsApp group where we can all communicate outside of meeting times so that any questions, updates and deadlines can all be discussed without needing to wait a week for a session and wasting time on the project, this has helped us keep on track and handle our time efficiently. Each member of the team completed their part of the project, which helped us work more efficiently throughout the project, therefore, work could be completed much quicker. The team dynamics were incredibly positive throughout the whole project, everyone has played their part by completing their bit and worked cooperatively, this allowed us to distribute the workload equally. Communication between every team member was really good as we would always talk through WhatsApp, so we'd be informed if anyone's missing a session, so that we wouldn't waste time on waiting for them and then filling them in over WhatsApp what they missed and what tasks they have been allocated. Overall, the positive team dynamics helped us work much smoother as we didn't have any disagreements as everyone supported each other and we just worked really well as a team, which allowed us to work more efficiently and reach the goals and meet the deadlines we have set.

Planning & Expectations

Due to the short time schedule, it was imminent that we discuss our expectations of the timeline. During our early meetings, we constructed a Gantt chart, using this we were able to comfortably finish one task after another. Below is our Gantt chart. Often, we found it easiest to do many of the tasks as a group, this meant we planned meeting and came together where a straightforward solution was not found. Otherwise, we compiled a list of tasks to complete and distributed 1-2 people to each task. Our concept selection was made up of us listing every goal and possible problems that these faced as seen in the image below, which resulted in around 3 ideas under every goal, this widened our perspective of things we could possibly create.



As a team we gravitated towards fire safety which fulfils Industry, Innovationa and Infrasturcture Goal. We generated a few ideas, considering various problems and their potential solutions, we were able to sort these into two subsections:

Fire Mitigation	Fire Suppression
Fire alarms Fire safe insulation Electrical fuse development Electrical break away/ timers Automated Household Fire Inspections Smart Stove which contains foam/suppression	Water fighting Drones Foam & inert powder bombs Lighter Fire gear Infrared visors / helmets Redesigning Fire helmets Sprinkler system

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The following week we individually researched current market gaps and found we wanted to construct a combined fire alarm with people counting systems. The market research was kept brief and straightforward in order to keep our goals clear.



Project st: Mon, 2/2/2026
Display w 1

SIMPLE GANTT CHART by Vertex42.com
<https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html>



Market Research

Objectives:

To understand our audience and demand of such fire alarms and systems. Using this to construct a specific target audience and consequently a sample plan. Our product aim: to create a commercial fire alarm with integrated people counters and safety systems, therefore market research must be conducted, comparing preexisting alarms and systems.

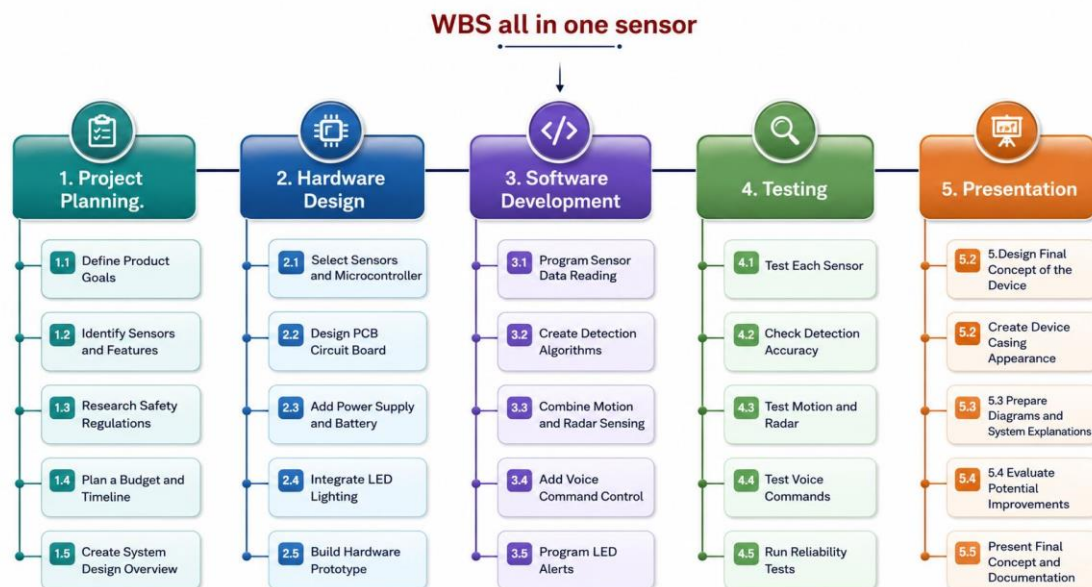
This report will consider the following of other companies: Technology, Battery, Disposal, Warranty, Weight, Recommended age, Fire alarm variations:

type	power	Purpose & features- direct quotes	warranty	Recommended age	weight
Smoke alarm	9V lithium battery	“Optical Smoke sensor for slow smouldering fires containing testing and silencing button. When it starts chirping it is a sign to replace battery. Smoke alarms are not suitable for kitchens and bathrooms this is because cooking fumes and steam can trigger the sensors and cause false alarms. It is important to clean alarms regularly to remove any particles which can cause false alarms.” (SB1-R optical smoke alarm 2026)	3	10	113
CO alarms	10-year battery pack	“Contains a test and silence feature. The Alarm Sounds at Level 85dB at 3 metres. Approvals and certifications: BS EN 50291-1:2018. Alarm monitors Carbon Monoxide Level” (Battery operated CO detector – 10-year battery 2026)	3	10	n/a
Heat alarms:	Lithium Battery	“Radio Frequency Heat Alarm provides effective heat detection. Featuring radio frequency interconnectivity making continuous monitoring possible and seamless integration with other safety devices in your system, like a hub. Containing Test and Hush Features: make easy maintenance and temporarily silence false alarms. Alarm Sound Level at 85dB at 3 metres. F1 rating” (Battery heat detector – 10-year battery & RF interconnectivity 2025)	5	10	n/a
Co and smoke	10-year sealed lithium battery. 3V	“Contains an Optical Smoke alarm Sensor and CO: Electrochemical CO alarm. With clear indicators and alerts to allow you to see which is present if the alarm sounds. Since it is an optical alarm fear not the steam and freely Install in further units and kitchen. DISPOSAL Waste electrical products and batteries should not be disposed of with your other household waste. Disposal the alarm and battery within the waste electronic and electrical equipment (WEEE) recycling scheme. To deactivate the alarm, use a screwdriver to remove the plastic pin on the back of the alarm. Insert the pin into the deactivation hole. Once the alarm has been deactivated, the battery will be depleted, and the alarm will stop emitting the low power warning chirp.” (SCB10-R combination smoke and carbon monoxide alarm 2025)			200g

Heat & Carbon Monoxide	10yr. 3 Volt. Rechargeable Lithium Battery Backup	“Advanced Heat Detection: Detects rapid increases in temperature, providing early warning of potential fire hazards before they escalate. Carbon Monoxide Detection: Monitors CO levels and provides timely alerts. Thermistor Class A1 detection – Alarm is triggered at 60oC. Carbon Monoxide Sensor: Electrochemical Scheme Rating: D1” (Combi Heat & Co Detector – Dual Protection 2026)	5	10	158
Heat and Carbon Monoxide (CO)	Built-in 10-year rechargeable Vanadium Pentoxide Lithium cells. Fully charged, the battery up to 6 months	“It holds a proven electrochemical CO sensor to detect the presence of toxic levels of Carbon Monoxide. Class A1 Thermistor sensor to detect dangerous levels of heat. The combination of CO and Heat detection makes it an ideal Alarm for kitchens, utility rooms and garages where fuel burning appliances maybe installed. Up to 12 Alarms can be interconnected – sold separately. The Ei3000 series is supplied with a mounting plate for quick installation. Each Alarm comes with built-in rechargeable backup batteries. Audio LINK+ app allows the user to download information from the Alarm. Alarm Sounder Piezoelectric Alarm. Plastic Material UL94V-0 flame retardant rated “ (EI3028 / 3018 alarms instruction manual)	10	10	350
Smoke & heat		“Battery powered smoke alarm which can be linked with up to 40 other Wiser Smoke Alarms. Smartphone notifications of detection and in-app status monitoring. Battery powered, no wiring necessary. Follows EN14604:2005 BS 5839 Battery operated Smoke Alarm - grade F1” (Battery heat detector – 10-year battery & RF interconnectivity 2025)			
Heat/ smoke Co	Main and 1x CR 2/3 AA seals battery	“Up to 1,700 m of wireless communication + with a range extender. Instant maintenance notifications, Visual alarm verification using MotionCam detectors or any NVR camera. Indoor/outdoor intrusion sirens activation- Built-in 85 dB siren LED sign of detector states. Smart Bracket panel to mount the device without disassembling the enclosure. Removable WAGO 221 connectors to secure the wires without a screwdriver. A lock to secure the device on Smart Bracket Pairing with a hub via QR code, with remote control and configuration in the app. Smoke sensor ignores vapor and water steam. Dustproof, impenetrable to insects, casing, does not require regular clean-ups The Haze Flow 2 software algorithm also protects against false alarms with the algorithm processing the data and confirm the alarm.” (Wiser smoke alarm - battery 2025)	2	7 & replaceable.	318

Smart Sensor 3C PC with Occupancy and People Counting	PoE (Power over Ethernet)	“Key Technologies: IoT Sensors: Monitors vaping, THC, air quality, chemicals, and hazardous gases. Audio Analytics: Detects aggressive behaviour, gunshots, and specific keywords like "help". Lighting Alerts: Features built-in multi-coloured LED lights for visual alerts, such as displaying red for danger. Privacy Compliance: The device performs all these tasks without recording any video or audio The 3C-PC model specifically uses privacy-protecting thermal sensors to measure the presence of body heat to find the number of individuals in a room.” (Avigilon Unity HALO-V3.00C-PC HALO IoT Smart Sensor 3C PC with Occupancy and People Counting.)	5 or 1 dep on were	n/a	365
Motion sensor	3v	“ Provides a 360° detection angle with a range of 4m diameter when ceiling mounted: 90° - 110° detection angle with a range of 5m radius when wall mounted Provides a 360° detection angle with a range of 4m diameter when ceiling mounted: 90° - 110° detection angle with a range of 5m radius when wall mounted” (Wiser motion sensor 2025)	1	5	n/a

To officially begin this project, we created a work break down structure, this allowed us to draw out the priorities of the project. We began to design the project and identify the critical path established by the tasks and their interdependencies.

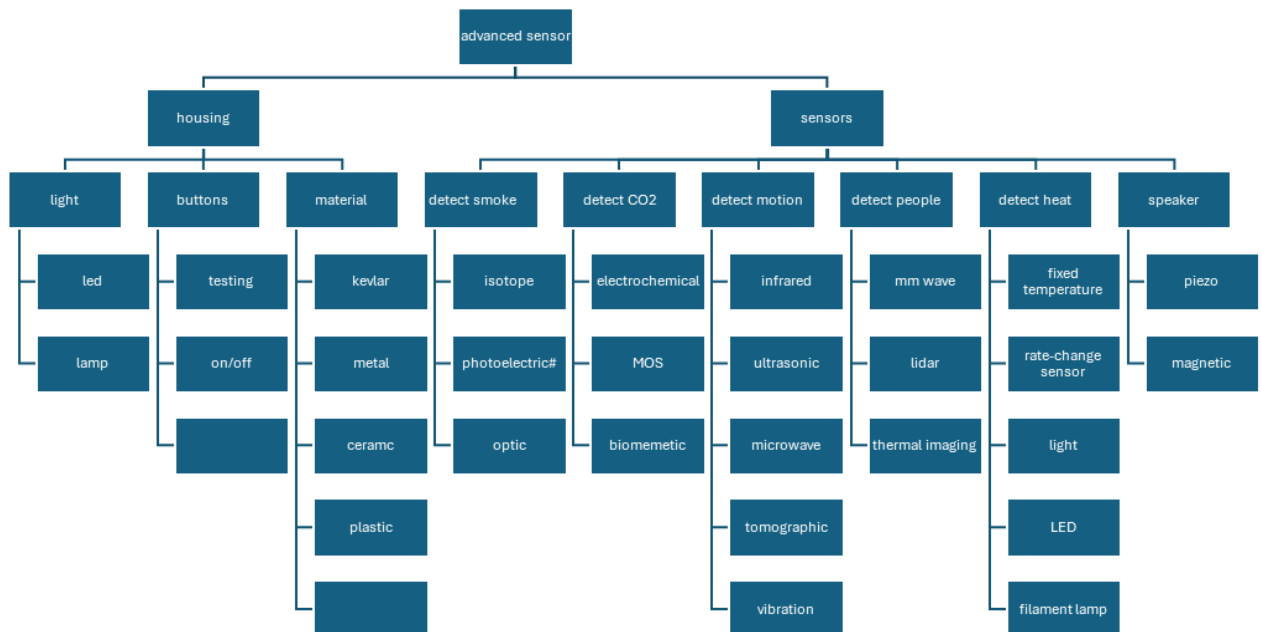


With any project a company faces’ risks. To prevent and mitigate accidents, our team constructed a Risk Register determining the major risks of our project.

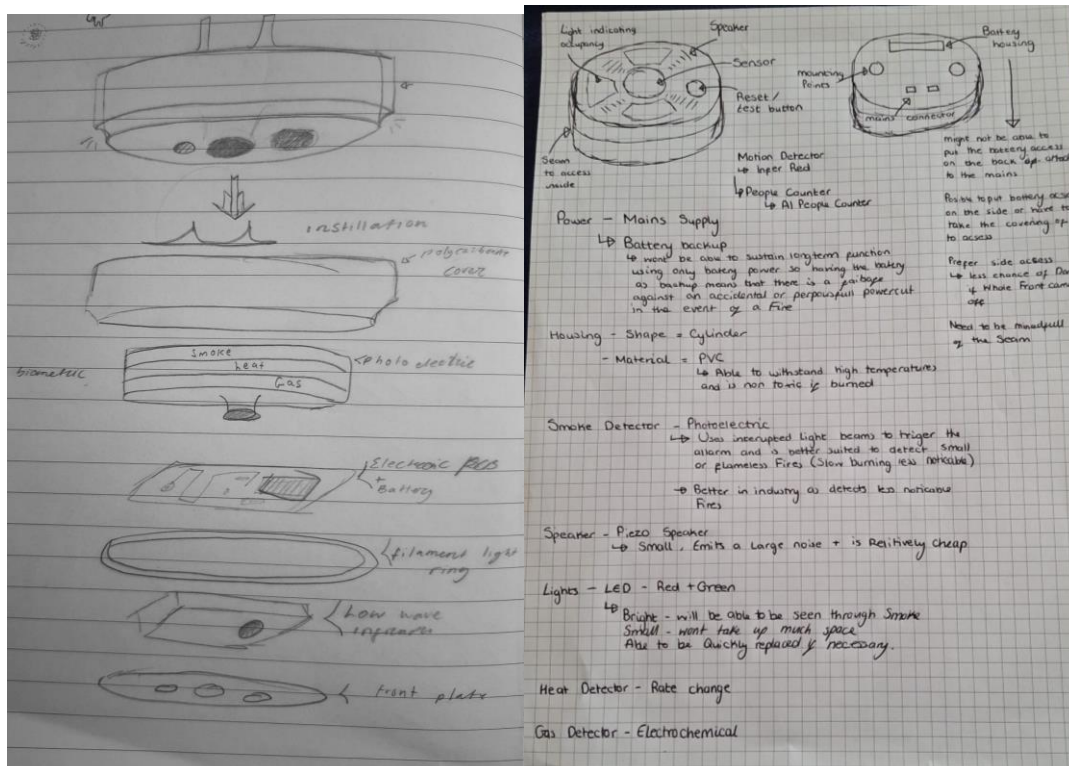
*Scales are on a scale of 1 to 10, 1 is close-to-insignificant and 10 is severe.

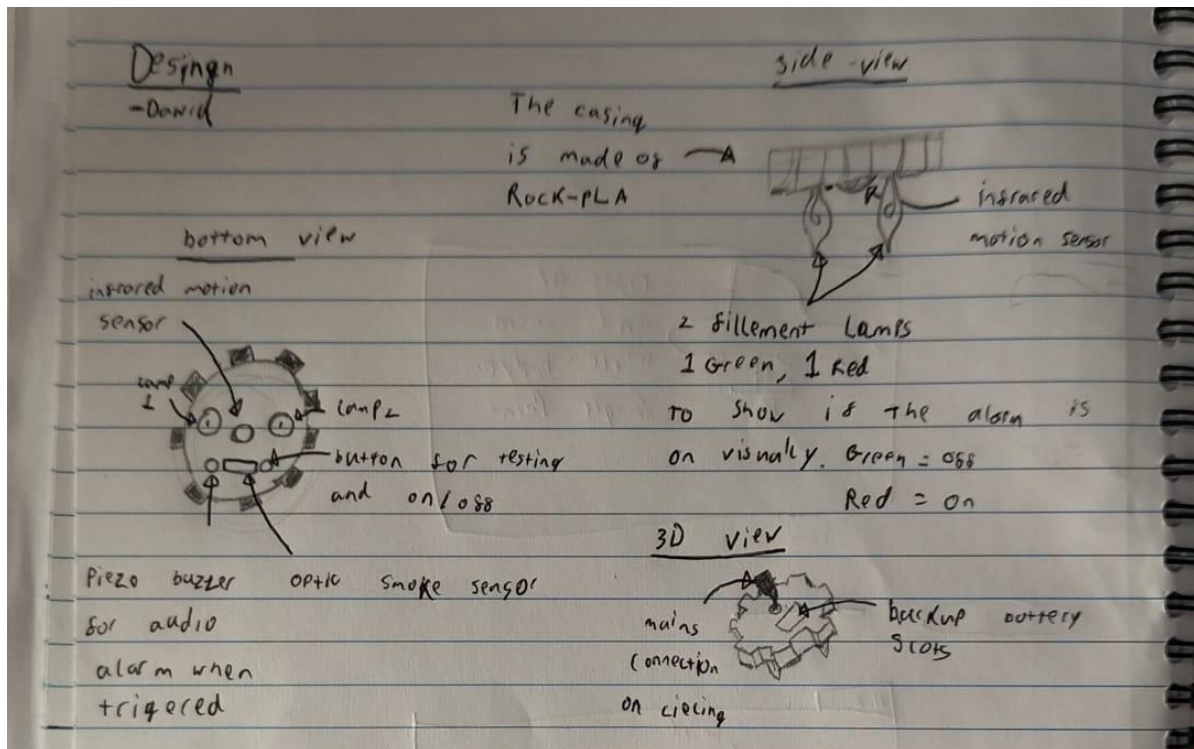
Risk Management Register									
Category	Description	Probability	Impact			Severity	Action	Owner	
			cost	schedule	mean				
supply chain	sensors arrive at different times to warehouse	6	4	3	3.5	4.75	Keep back stock of each sensor	MT	
supply chain	housing arrives late or broken	3	5	8	6.5	4.75	Add penalty to contract (shipping or manufacture)- Confirm delivery	MT	
supply chain	housing arrives deformed or faulty	4	5	6	5.5	4.75	Add penalty to manufacturer's contract, contact about fault	MT	
supply chain	sensors arrive faulty	2	7	9	8	5	Add penalty to Sensor Company Contract	MT	
supply chain	sensors arrive broken or late	2	6	6	6	4	Add penalty to contract (shipping or manufacture)- Confirm delivery	MT	
human	Electrical shock from internal battery	2	4	4	4	3	ensure battery is enshruded and properly use risk signalling	MT	
human	Incorrect installation	5	2	3	2.5	3.75	ensure installation instructions are clear and concise	MT	
human	incorrect operation	4	2	4	3	3.5	ensure operation manual is detailed and explanatory	MT	
human	harm in manufacturing process	4	3	6	4.5	4.25	ensure proper PPE is worn and each team member is trained	MT	
human	failure to maintain system	3	3	4	3.5	3.25	make sure scheduled maintainance is completed and logged	MT	
financial	faults in our product	3	8	9	8.5	5.75	Ensure we properly vet and test thoroughly to mitigate costs later on	MT	
financial	Faults in components	2	8	9	8.5	5.25	secure long term supply deal at a fixed price	MT	
financial	high production cost	7	7	2	4.5	5.75	mitigate costs where reasonable, offset production cost with product price	MT	
financial	change in shipping costs	8	3	5	4	6	create and maintain financial buffer to offset changes in prices	MT	

Design Criteria and Specification



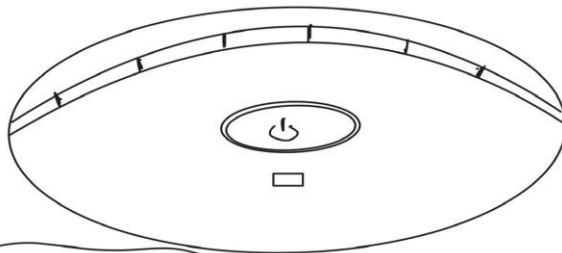
In order to choose a final design, we needed to consider all the functions of the product, therefore, we created a function tree to show all the requirements and all the options of technology we had. This proved to be especially useful as it specified and narrowed down all the options. Combining the diagram and our discussions we had a lot of considerations and intricacies. We decided to each come up with our own individual rough sketch of an alarm with all these functions. This way we could examine more possibilities and then settle on the best features of each of our ideas.





Multi-Sensor

Made with Goodnotes



LED key

- - Normal
- - Alarm
- - Testing
- - Fault

features

- Isotope (smoke detection)
- Allos (gas/co detection)
- Infrared (Heat detection)
- Heat sensor (Temperature)
- Piezo speaker (sound)
- Button (on/off, testing)
- Ceramic casing (case)
- Battery (Back up),

Description

- Round, flat ceiling mounted ceramic casing unit.

Pros/cons

- + multi-sensor system gives high accuracy and early detection
- Complex and regulated design (expensive to manufacture).



Filament bulbs	
Pros	Cons
Cheap	Low heat tolerance
Easy to install	Glass is dangerous when broken
Common and easy to source	Low light level relative to same size LED
	Protrudes
Magnetic speaker	
Easy to source	Expensive
Good sound range	Low heat stability

For our design specification, we made sure to think deeply and evaluate each area in which we would need to have a realistic requirement. These requirements are as follows:

Safety

- We must make sure there's no exposed electronics, as any case of this could harm anyone installing, repairing, or maintaining the system.
- We must make sure that the entire system is up to safeguarding standards, even in the event of an emergency we need to maintain a proper safeguarding standard. This includes making sure we aren't scanning or saving anything that could identify people because of privacy laws, and in workplaces can go against labour laws to do with employee monitoring. And since we are selling commercially, this is highly important.
- Due to this system including both a battery and a mains connection, in the case that the mains connection is severed or damaged and the system is running purely on backup power, the system will prioritise lower energy sensors and alarms instead of the entire system, allowing it to stay active for longer periods and able to save more lives.
- We must make sure that this entire system is up to the correct legislation and regulations wherever they are relevant, as since this is a safety system any issue with regulatory bodies means the entire thing could be forced off the market.

Quality

- The entire system must be ruggedized, this is so the components and entire system is safe to manipulation, user handling errors, and since this is going to be used within hazardous situations it needs to be resilient and overly redundant.

- Since fires can cause hazardous materials and gasses to be created, therefor the entire system needs to be able to survive and be active in an environment where these materials and substances are present.
- Making the entire system able to resist open flame and extremely high heat offers the same benefits as ruggedization, where the system will be able to work in the environment that it's designed for. Also, if we add a large redundancy factor to the heat it can withstand, it ensures that even the system will always be able to survive situations it would realistically endure within its own lifecycle
- Since the system is planned to basically be dormant for the majority of its life cycle we need to ensure that the entire system is able to be left alone for long periods of time without maintenance and repair, but during these long periods of time is still able to maintain its effectiveness and efficacy.

Economical

- Since this is a system intended for sale, and purchase by commercial consumers, it is important that we keep this system within a realistic price range for its purpose. It also needs to be competitive, as since this system is comprised of commercially available sensors, we need to maintain a price point that isn't too much higher than the sum of our parts.
- This system also needs to be energy efficient for a few reasons. One of which is that since this is a product to be sold, it needs to be feasible within a commercial budget and if the energy efficient is good enough, it will be more attractive to businesses. Another reason is that since this needs to be able to continue activity on its own backup power, the more energy efficient the system is the longer it's able to maintain itself on its own backup power.

Convenience

- The system needs to be easy or at least accessible to repair. The system is intended for use within large areas and as many businesses as we could realistically sell to, and therefor if it were only able to be maintained and repaired by our own technicians, it would be a completely unrealistic and unattractive idea for a working business. For this same reason we need to make sure the system is easy to reset in the rare case that it falsely activates.

Sustainability

- We need to make sure that the system is made using sustainable materials, and if we can't source sustainable materials, we need to make sure that its lifecycle is long enough and that its maintainable and repairable enough to where its chance of being thrown in landfill or damaging the environment is absolutely minimised.

Power Consideration

The fire alarm requires a stable voltage. It will be a low voltage, as it doesn't need much to operate all components safely and reliably. The system uses a piezo buzzer, an RGB LED, motion sensor, CO2/MQ gas sensor, heat sensor and a photoelectric sensor. Most of these components either run on 3.3V or 5V DC supplied by the circuit, in the prototype it would be an Arduino.

The buzzer requires enough current to produce a loud warning sound during an alarm and has different frequencies depending on the alarm triggered such as CO alarm or the photoelectric sensor. Additionally, the led will be flashing. Green when there's nobody in the room, red if there's people in the room. It uses additional energy as it lights up when active and operates at the same time as the other components.

The motion sensor requires continuous power operation when the alarm is turned on so it can constantly monitor for movement and detect people. The CO2 gas sensor has one of the highest energy requirements as it contains a small heating element that must stay powered for the gas detection to be working properly. The heat sensor and photoelectric sensor use relatively small amounts of energy as they mainly monitor environmental changes and send signals directly to the controller. This does not require much power.

For continues use, the system should use a 5V DC power source with sufficient current capacity to support all sensors and outputs operating at the same time in case of an emergency, even if the likelihood that all sensors are active at once is low. A battery backup aside from the supply will also be required to ensure the fire alarm continues functioning during a power outage.

Because the fire alarm system uses a relatively small amount of electrical energy as most of the components are low power sensors so the system running on a 5V power supply will use around 2–5 watts of power during normal operation while off, depending mainly on the CO₂ gas sensor since it contains a heating element that draws power continually. For one full day of operation, the system would use approximately 0.05 to 0.12 kilowatt-hours (kWh) of energy because of it continues powered on state for the sensors, despite its dormant state. Over the course of one year, this would equal roughly around 18 to 44 kWh.

When the alarm is activated, energy usage temporarily increases as the buzzer and RGB LED operate along with all the other sensors. However, alarms are only expected to run for short periods, so they have little impact on the total energy consumption, making almost no difference to cost of running the fire alarm year-round. Overall, the system is designed to remain energy efficient while still providing continuous monitoring and reliable fire detection.

Concept Selection and Development

We compared all our designs together, as discussed, the merits and drawbacks to each feature of each design. We cross-referenced this all to our requirements of the product and so we came to the conclusion to use:

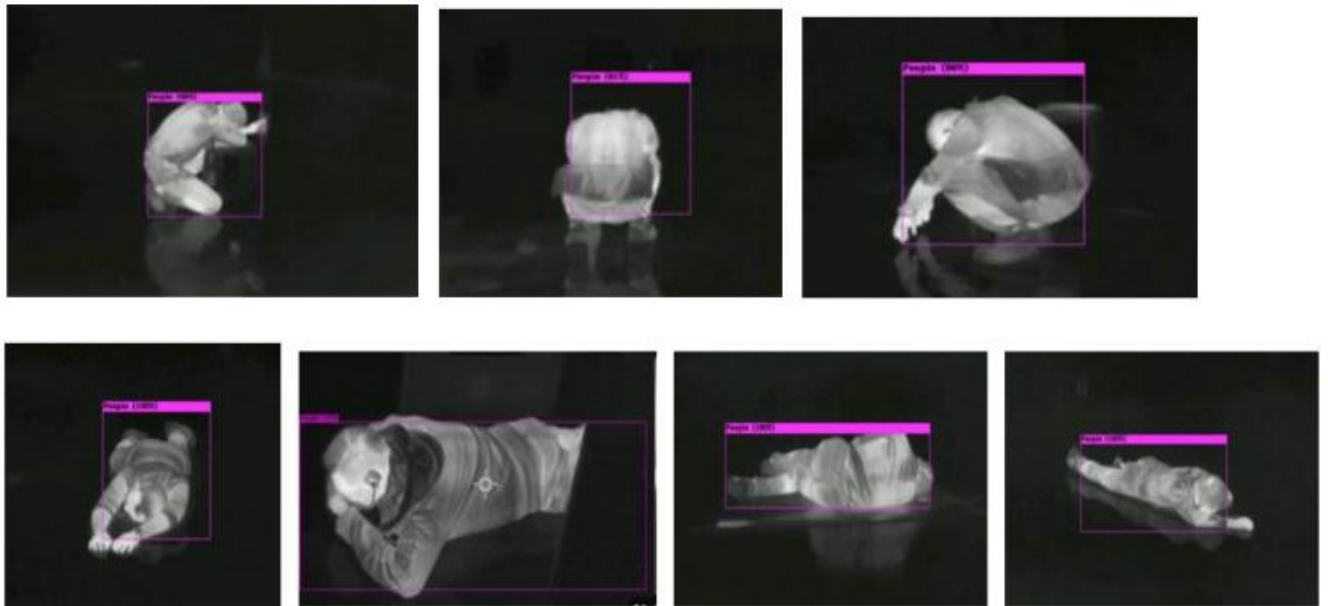
The casing being a simple, ergonomic semi-circle shape, with no sharp edges or corners. We chose this to allow the preservation of aesthetics within any room it is installed in. This also allows for more space for all the components to fit inside. This casing in a grey colour to fit any room without looking like an intrusive camera or sensor. Made out of Fire-retardant PVC as it is robust, cheap and still maintains all fire alarm standards

Surrounding the casing, a ring LED instead of a single bulb, to light up the entire room red if the occupancy level is zero or red if the occupancy level is not. We thought it vital to emphasise safeguarding by constructing a secondary- relay circuit, preferably with a bridging compound to ensure that the people counting sensor is only ever on if the alarms trigger. A simple Piezo buzzer to buzz once the alarm is triggered and a speaker to communicate the occupancy. An MOS gas sensor since it can detect various harmful gases. A testing and silencing button for false alarms or to check that the alarm works. A chirping signal if the battery were to be low would also have to be implemented.

We have also concluded to have a 5-year warrantee and a 10-year life period of our devices, to demonstrate our interest in the safety of our customers and the efficiency of our product.

After doing lots of research for deciding a people counting sensor, we came across an extremely relevant and useful research paper on the effectiveness of low-wave infrared cameras in fire scenarios. This paper concluded that they were in fact certified to be used in professional fire response teams. (Tsai et al., 2022) describes the precision at which people were identified with, and the accuracy achieved by running this system through deep learning.

The images below are from the same study, where the model run was able to consistently able to identify crouched or collapsed figures in smoky/ low visibility environments.



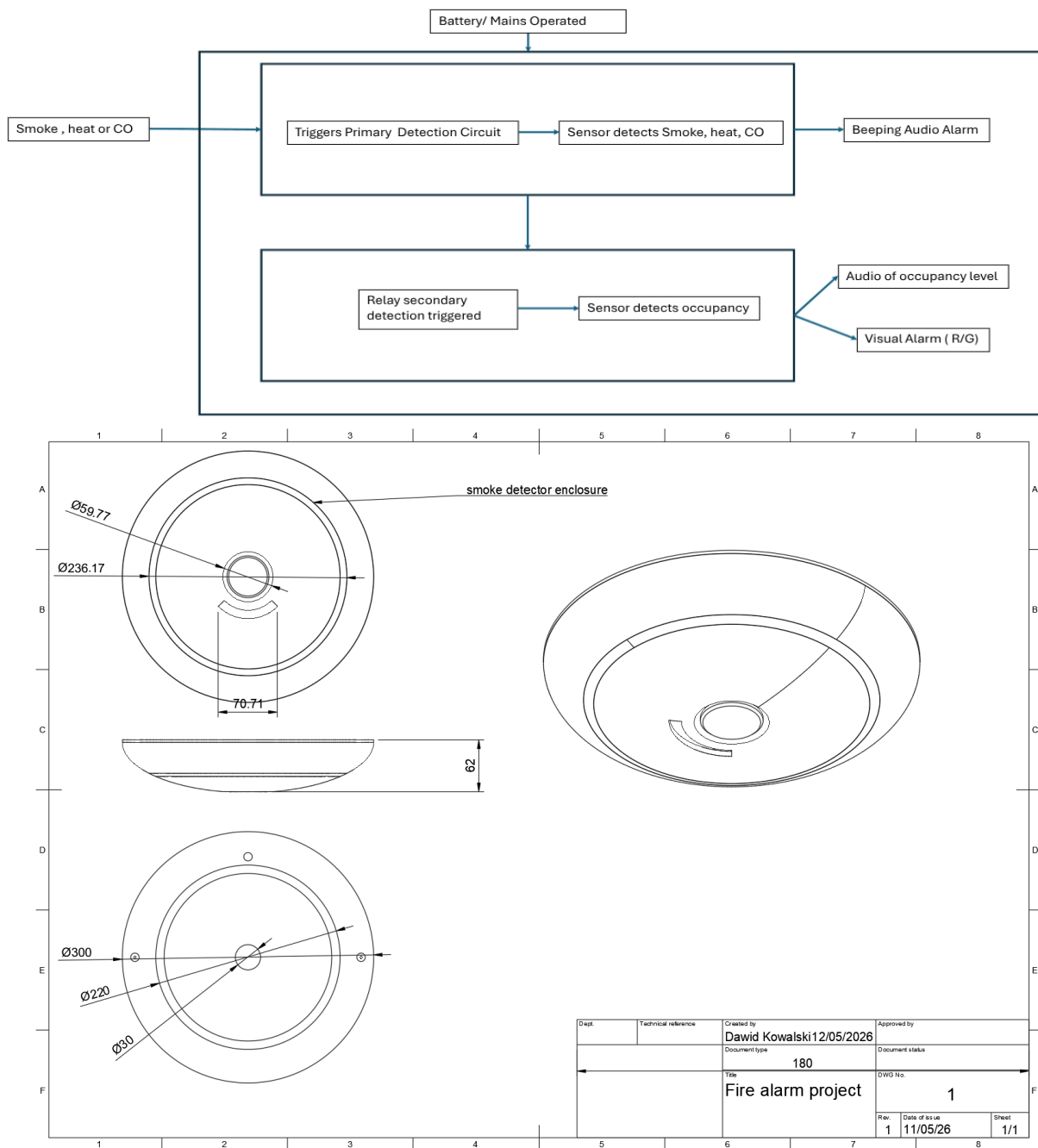
IMAGES

FROM:

Tsai, P.-F., Liao, C.-H. and Yuan, S.-M. (2022) 'Using deep learning with thermal imaging for human detection in heavy smoke scenarios', *Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios.*, 22(14), p. 1. doi:10.3390/s22145351.

Therefore, we decided to settle on a similar low wave infrared camera for the alarm.

From there, we also needed to specify how we wanted the product to function, together we constructed a function tree/ diagram.

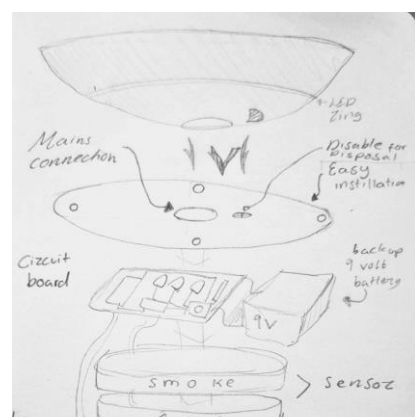


Eventually we came to produce these designs as seen above.

QDF

QDF – Quality Function Method is used to relate customer requirements into a scale suitable for engineers. The House of Quality quantifies the importance of customer needs against the technical characteristics.

Customer Requirement	Importance (1–5)
Reliable fire detection	5
Accurate people counting	4
Low false alarm rate	4
Fast emergency response	5



Low power consumption	5
Easy maintenance	3
Compliance with safety/privacy laws	5
Long operational lifespan	4
Cost-effective system	3
fire-resistant casing	4

House of quality matrix, correlation of customer and product

Strong = 9, medium = 3, low = 1

Customer requirements	Sensor accuracy	Thermal resolution	Response time	Power use	Battery backup	Alarm output	Fire R=resistance	Lifespan	Maintenance	Cost
Reliable fire detection	9	3	9	1	3	3	1	1	1	1
Accurate people counting	9	9	3	1	1	1	1	1	1	1
Low false alarms	9	3	3	1	1	1	1	1	1	1
Quick emergency response	3	1	9	1	3	9	1	1	1	1
Power efficiency	1	1	1	9	9	1	1	3	3	3
Easy maintenance	1	1	1	3	3	1	1	3	9	3
Safety/privacy compliance	3	3	1	1	1	1	3	1	1	1
Long operational lifespan	1	1	1	3	3	1	3	9	3	3
Cost-effective system	1	1	1	3	1	1	1	3	3	9
Fire- retardant & durable casing	1	1	1	1	1	1	9	3	1	1

Technical importance ratings

Engineering Specification	Total Score
Sensor accuracy	146
Detection response time	118
Thermal sensor resolution	86
Power efficiency	78
Battery backup lasting	74
Fire resistance rating	67
System lifespan	65
Maintenance interval	57

Manufacturing cost	54
Alarm output	52

Material list:

Battery: Panasonic Pro- 9v battery

Price: £1.72 - £9.99

Lithium-ion Battery,

Lasts 10 years roughly matching typical fire alarm lifespan

Needs recharging roughly every year

Retain charge when not in use

Rechargeable so good for the environment

We chose lithium-ion rechargeable batteries over other kinds is that not only are they more affordable they are non-toxic unlike NiCd batteries as well as having benefits when it comes to recycling them with lower emissions than new mining meaning that they are more environmentally friendly. They also have a lower self-discharge compared to NiMH and NiCd batteries.

Speaker: Murata PKM13EPY H40 00-A0

Price: £1.20- £1.60 per unit

Piezo Buzzer

Very low current draw and long operating life

Voltage Range = 3-20V

Reasonable Frequency = 4kHz

Sweet spot for hearing (can be loud without significant power draw)

Uses lead free ceramics- RoHS compliant

Long operation life- reduces component waste

We chose to use piezo buzzers over other sound producing devices as they use less materials so are not only are they cheaper to use but they are more environmentally friendly as they use simple materials, have a low energy use and typically a long life which aligns with other components of the alarm meaning there is less need to replace the unit. They are also more reliable due to the lack of moving parts and so are less likely to require maintenance to fix issues.

Casing: Simplas PVC recycling – will have to find someone to produce casing

Tubeway Extrusions- will produce the casing

Recycled PVC

Able to withstand high temperatures

Nontoxic when burns

Environmentally friendly while maintaining the necessary integrity needed during a fire

We chose to use recycled PVC over other materials since it not only has a lower environmental footprint, but the fact that fewer harmful toxins are realised when manufactured. Another reason why we chose to use recycled PVC is the lower material costs, so it is cheaper to mass produce as the casing is the thing to be most damaged in the event of a fire by either heat or smoke. There is also the fact that if the casing were to catch the recycled PVC does not produce any toxic fumes when burned that could potentially make the situation worse.

Infer-red sensor: Emberion Ultra-Low-Cost SWIR Sensor

Price: production cost < €50 in volume

Wavelength range = Short-wave IR (SWIR), typically 400–1700 nm

Environmentally friendly — uses quantum dot materials and wafer level packaging, reducing waste and energy use

Is reliable and uses low power so won't drain system when active

We chose to use infer-red sensors over other kinds of motion sensors due to their compact nature as well as their compatibility with people counting software. Another reason as to our use of infer-red sensors is the fact that they have a lower power consumption as well as being a non-contact method so there is less chance of breakage as well as being resistant to corrosion. They were also selected due to their reliable detection as well as good and consistent performance over time which will limit the number of false alarms triggered.

Photoelectric sensor: XP95 Smoke Detector

Price: £22-£35 per Head

Uses infer-red light chamber so there is no radioactive material used

Has a recyclable housing

Is EN54-7 certified and is long lasting so it reduces component waste

Requires low power to run

We chose to use a photoelectric sensor as the smoke detector for the alarm this is because it uses interrupted light beams rather than radioactive materials like some others. This is more environmentally friendly. We also chose to use this type of sensor as it is more reliable when responding to smouldering fires and is less susceptible to steam and humidity giving a more consistent performance and reducing the risk of it being a false alarm. The photoelectric sensors also perform better in real world environments and offer the strongest balance of eco-friendliness, affordability and reliability.

MOS gas detector: SGX / e2v MiCS Series

Price: £10–£20 depending on distributor

Detects a range of gasses (O₃, CO, VOCs, NH₃, NO₂.)

Designed for environmental monitoring & is highly reliable

Is suitable for industrial applications

Uses ceramic semiconductors and has low power requirements

We chose to use MOS gas detector as they are not only lightweight, but they are robust and long-lasting. They are also lower in cost while also still detecting a wide range of gasses that may be produced in an industrial setting. MOS gas detectors also have lower power consumption rates and are highly stable. They also are highly sensitive and reliable having quick response times being able to detect low levels of gas accurately.

Ethical Consideration

This product has two significant features that must be considered prior to the development of this product.

- Efficacy of a fire alarm containing a Low Wave Infrared Sensor since it could breach safeguarding concerns
- Material, component selection

Material & Component Selection

At Pufferfish Security, we do our best to ensure that our product does not harm the environment. For that reason we chose to use recycled PVC over other materials due to the fact that it not only has a lower environmental footprint, but the fact that fewer harmful toxins are realised when manufactured. There is also the fact that if the casing were to catch the recycled PVC does not produce any toxic fumes when burned that could potentially make the situation worse. By selecting recycled PVC, we have considered the lifecycle of our product and its process of disposal.

We also chose lithium-ion rechargeable batteries as when it comes to recycling them with, they release lower emissions than new mining meaning that they are more environmentally friendly. They also have a lower self-discharge compared to NiMH and NiCd batteries. Therefore, if the average lifetime of 10 years somehow does discharge before, the batteries are easily rechargeable with a regular mains adapter.

Furthermore, it is vital to understand that every fire safety sensor is by legislation recommended to be replaced every 10 years. Therefore, the team has designed the fire alarm to have a replaceable attachment which not only saves money of the customer but also the environmental impact. We encourage our customers to send through any of our outdated alarms for us to recycle and reuse in exchange for a significant discount for further models.

Low Wave Infrared Sensor

All sources listed for this are listed in references.

Employee monitoring:

Some issues that occur with using monitoring devices in the workplace is employee rights. As the alarm contains a sensor to monitor whether people are in the room this creates a series of issues regarding employee rights. As stated in the (Ciocan, 2024) monitoring of employees is allowed if it follows a set of criteria. As the sensor is only going to record the occupancy of a room in the case of a fire many of these criteria are not applicable. However few criteria regarding employee monitoring refer to monitoring the positions of workers. While these are mainly focused on monitoring employees for productivity and ensuring that such data isn't used against them a similar approach can be used in reference to the alarm due to the monitoring component. Due to the guidelines the monitoring of employees is allowed if it is proportionate to the need and that employees are aware of just how they are being tracked, in what situations and are informed of any data that is stored or shared from this process.

these issues can be combatted by informing employees of the monitoring and make it clear that it only comes into effect in the event of a fire. We can further prevent issues by ensuring that if cameras are used to identify people in the room that recording or any information gathered is not stored nor shared with an external source. This will ensure that there are no issues due to the data protection acts. Another way that we can prevent issues is by having employers where the devices are going to be installed to perform a Data Protection Impact Assessment (DPIA) to further insure that their employees' rights and privacy won't be violated as well as ensure that no data is compromised due to the nature of the work (e.g. client info in a lawyers or doctor's office) this is more likely to be an issue if cameras are used to determine if people are present in the room compared to if alternate detection devices and methods are used.

Social:

Some of the main ethical concerns with cameras are privacy as well as autonomy. One way that cameras effect privacy is that they can be used to intrude in people's personal space. This can create what is known as the "chilling effect" that causes people to alter their behaviour due to the presence of monitoring. This could affect people who are in the presence of the alarm as despite disclosure on the purpose of the alarm, people may believe that they are being monitored which could cause emotional distress if they believe that their actions would be punished due to the monitoring.

One way to combat these privacy issues is to ensure that the alarms are only placed in necessary areas (Long, 2023) even going as far as to alter design to perhaps remove the monitoring component in sensitive areas such as restrooms, changing rooms and other areas where cameras are prohibited. Ethical guidelines on cameras emphasize transparency, informed consent, and proportionality, meaning surveillance should be limited to what is necessary and regularly reviewed for effectiveness.

Another ethical issue is data misuse. If the design where to include a camera to identify if people are still in the rooms unauthorized access, sharing, or manipulation of footage are issues that would need to be addressed as it could lead to discrimination, exploitation, or privacy violations. (Young, 2025) This could be combatted by

ensuring that the camera is a closed system that cannot be accessed or monitored by external methods and that if any footage is obtained that it is not stored or transmitted in any way. However, if the sensor doesn't include a camera some of these issues won't apply however steps would still need to be made to ensure that privacy and data is protected.

Legal:

On the Legal side privacy laws and consent requirements are critical. Organizations must inform individuals about surveillance, obtain consent where necessary, and limit data collection to legitimate purposes (*Surveillance Camera Code of Practice pursuant to the Protection of Freedoms Act 2012* 2023). (e.g. ensuring that the sensor only activates in the event of a fire, that no data gathered by the sensor is recorded and anyone working in proximity to the alarm is informed of how it works and when it is active).

Data security and retention are also legally mandated. Organizations must protect captured footage from unauthorized access, define retention periods, and establish protocols to maintain confidentiality and integrity. However, this is not much of an issue as the alarms shouldn't have to retain footage as the primary purpose regarding the sensor is to alert if anyone is in the room in the event of a fire and so footage isn't required. Failure to comply with these regulations can result in legal penalties and loss of public trust.

Some good practices to help combat these issues:

- Transparency and consent: Clearly inform individuals about monitoring and obtain consent where required.
- Proportionality: Use cameras only to the extent necessary for safety or operational purposes.
- Data protection: Implement encryption, access controls, and secure storage.
- Regular review: Continuously assess the effectiveness and necessity of surveillance measures.
- Ethical oversight: Ensure policies respect dignity, autonomy, and human rights, particularly in sensitive environments like homes or workplaces.

Rules about monitoring devices placement: In the UK, the placement of monitoring devices is regulated by several laws to protect individuals' privacy rights. Some of these rules are as followed:

- Privacy Rights: CCTV cameras must not capture private spaces without consent, as this can infringe on individuals' privacy rights.
- Public Areas: Recording within one's own land is generally straightforward, but capturing footage of neighbouring homes or public spaces requires additional legal responsibilities.
- Clear Signage: Clear signage is required for public areas or other people's property, as verbal notice alone is not sufficient under UK law.
- Sensitive Locations: Filming sensitive locations like schools or hospitals without permission can lead to legal action.
- Data Protection: Ensure compliance with the Data Protection Act 2018 and the UK GDPR, which govern the processing of personal data captured by CCTV.

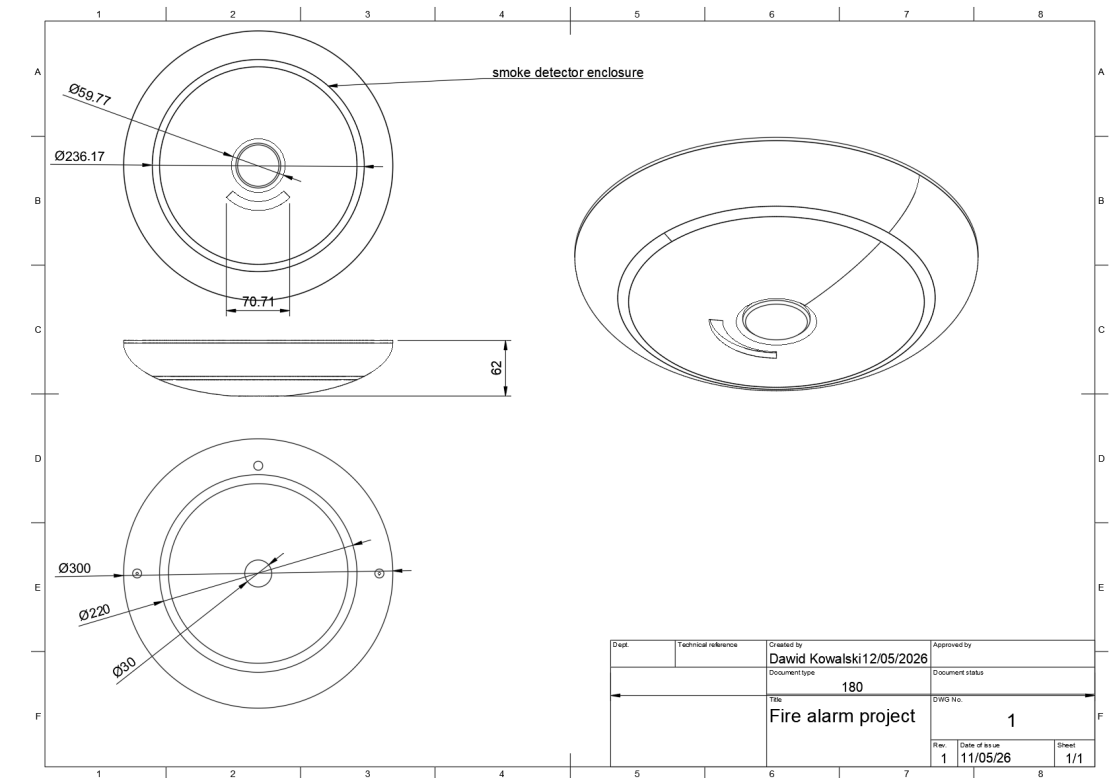
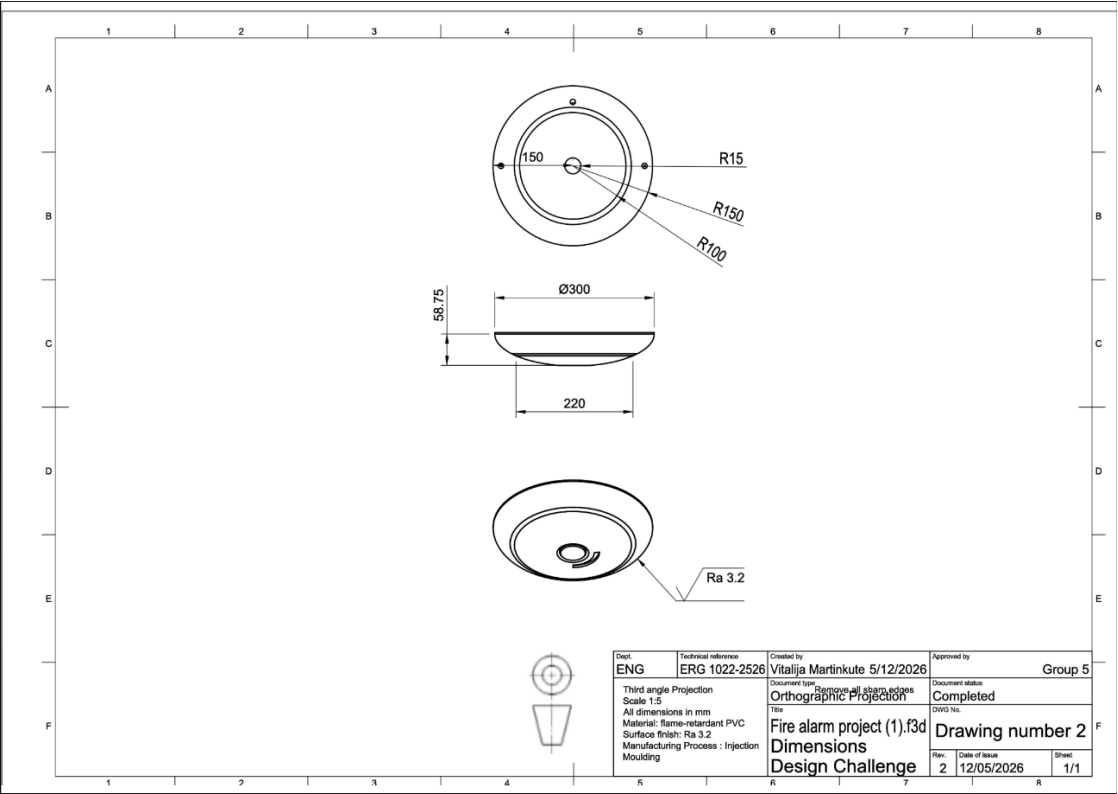
While these rules and guidelines are primarily for CCTV cameras some of the rules can be applied to the alarm. Such as the privacy rights (if a camera is used over an alternative sensor), sensitive locations (again if a camera is used), and data protection (can be applied to any kind of monitoring device not just cameras as there is multiple points of data that could be gathered).

Concept Feasibility

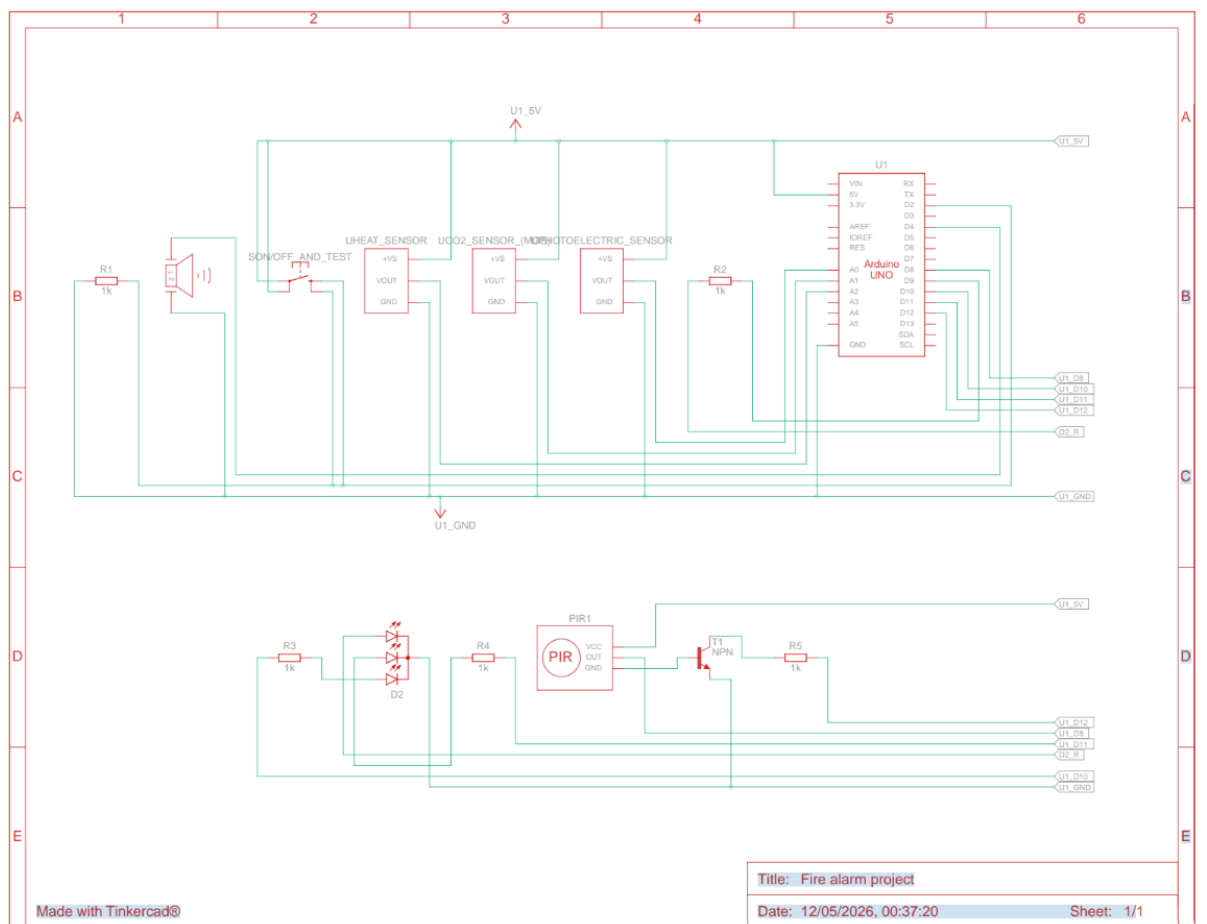
This project is generally feasible since most of the components in the sensor such as the heat detector or the infra-red light are already in circulation and are easily accessible to be integrated into most systems. This means that the real problem behind this isn't to make modern technology it is instead integrating all the part into one efficient

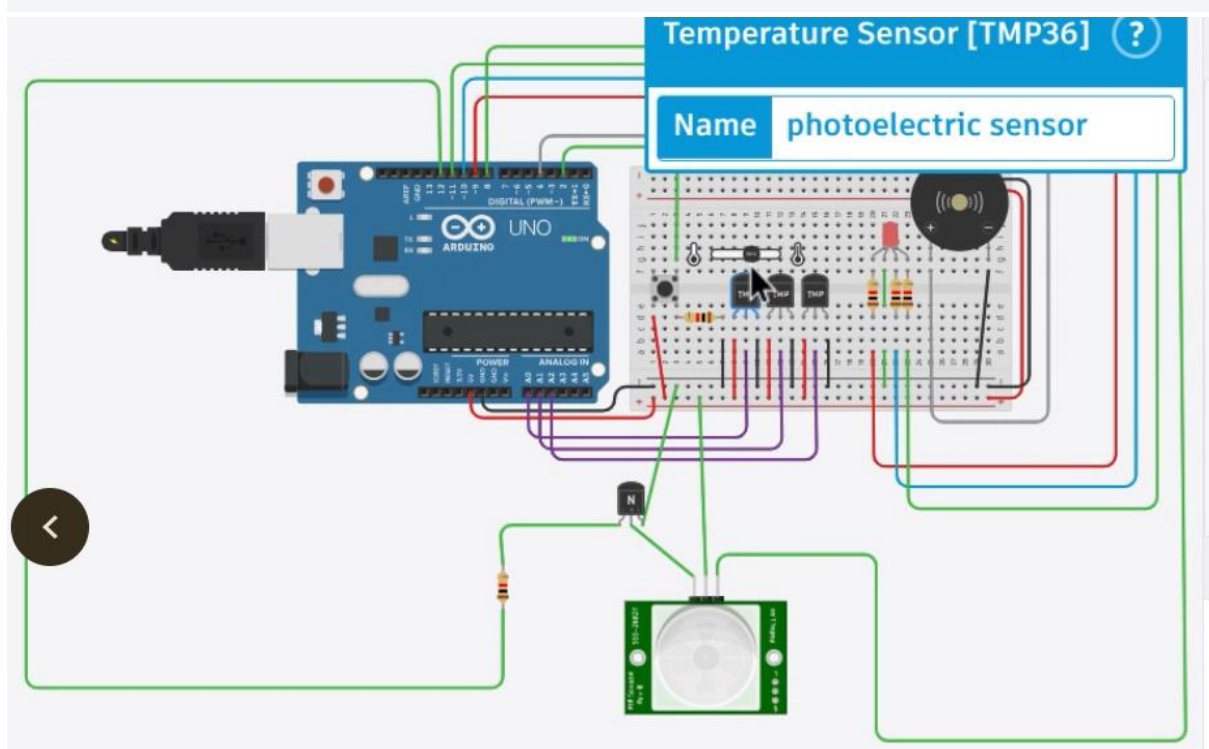
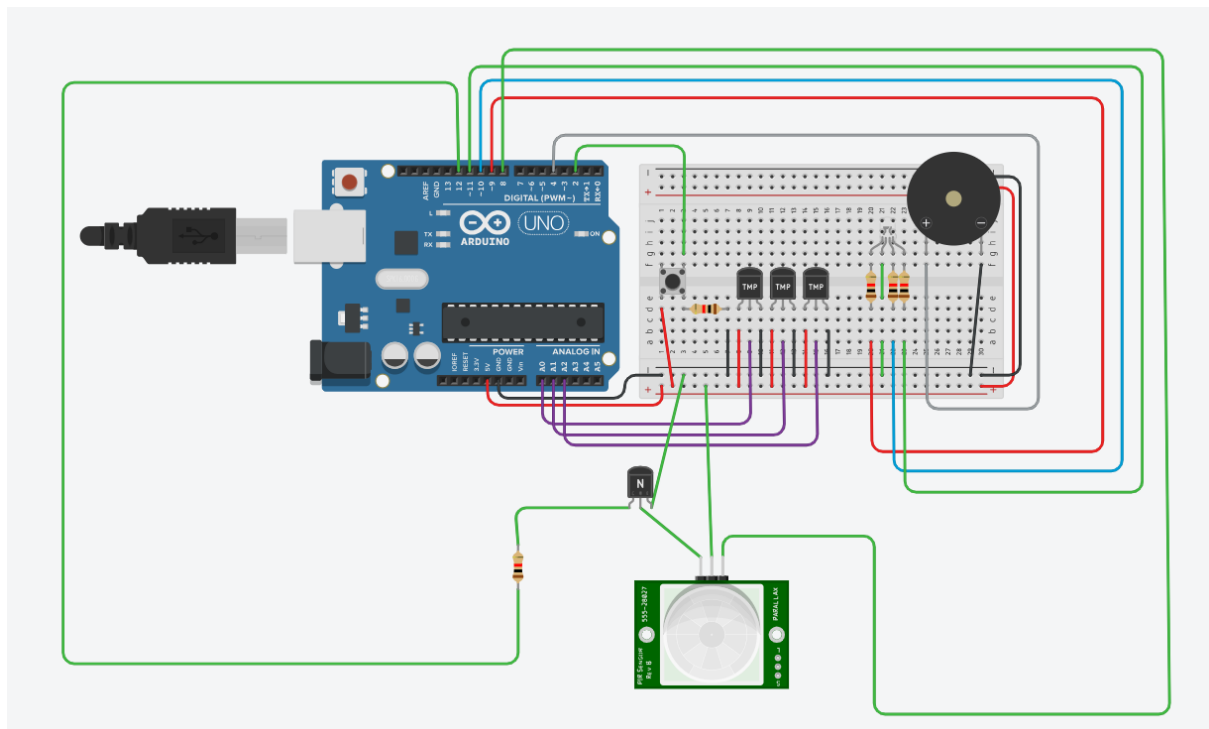
alarm. However, the project becomes more unrealistic if we add to many different components into one alarm as that could trigger false alarms and the overall complexity of all the sensor in one big sensor could make it extremely difficult to make so there needs to be a balance. Overall, it is realistic if there aren't too many components and the project is carefully managed.

Detailed Design









```

Text
1 // Fire Alarm System - TMP36 analog sensors version
2 // TMP sensor 1 (photoelectric substitute): A0
3 // TMP sensor 2 (CO2/MOS substitute): A1
4 // Button (manual/test): pin 2
5 // LED red pin 9
6 // LED green pin 11
7 // LED blue pin 10
8 // Buzzer: pin 4
9 // People counter switch: pin 8
10
11 //Pin Definitions
12 //const int PIN_LED = 3;
13 const int PIN_LEDRD = 9;
14 const int PIN_LEDBLUE = 10;
15 const int PIN_LEDGREEN = 11;
16 const int PIN_BUZZ = 4;
17 const int PIN_BUTTON = 2;
18
19 const int PIN_TEMP1 = A0; // TMP36 substituting photoelectric sensor
20 const int PIN_TEMP2 = A1; // TMP36 substituting CO2/MOS sensor
21 const int PIN_TEMP3 = A2; // TMP36 being the heat sensor
22 const int PIN_PIR = 8;
23 const int PIN_SECONDARY = 7;
24
25
26
27
28 const float TEMP_THRESHOLD = 50.0;
29
30
31 //Thresholds
32
33 //Timing
34 const unsigned long HOLD_THRESHOLD_MS = 2000;
35 const unsigned long TEST_DURATION_MS = 5000;
36 const unsigned long FLASH_INTERVAL_MS = 300;
37
38 // State
39 bool alarmActive = false;
40 bool manualAlarm = false;
41
42 int peopleCount = 0;
43 int lastPeopleSwitch = LOW;
44 int lastButtonState = LOW;
45
46 //NEW CODE HERE
47 bool sensorActive = false;
48 bool sensor2Active = false;
49 bool sensor3Active = false;
50
51
52 unsigned long buttonPressTime = 0;
53 unsigned long lastFlashTime = 0;
54 bool flashState = false;
55
56
57 // functions
58 float readTemp(int pin);
59 void triggerAlarm();
60 void clearAlarm();
61 void runFlash();
62 void runTestMode();
63
64 void setup() {
65     pinMode(PIN_PIR, INPUT);
66     pinMode(PIN_SECONDARY, OUTPUT);
67
68
69
70 digitalWrite(PIN_SECONDARY, LOW);
71
72
73 pinMode(PIN_LEDRD, OUTPUT);
74 pinMode(PIN_LEDBLUE, OUTPUT);
75 pinMode(PIN_LEDGREEN, OUTPUT);
76 pinMode(PIN_BUZZ, OUTPUT);
77 pinMode(PIN_BUTTON, INPUT_PULLUP);
78
79 analogWrite(PIN_LEDRD, 0);
80 analogWrite(PIN_LEDBLUE, 255);
81 analogWrite(PIN_LEDGREEN, 0);
82
83
84
85
86 Serial.begin(9600);
87 Serial.println(" Fire Alarm System Ready");
88
89 }
90
91 void loop() {
92     int buttonState = digitalRead(PIN_BUTTON);
93
94     float temp1 = readTemp(PIN_TEMP1);
95     float temp2 = readTemp(PIN_TEMP2);
96     float temp3 = readTemp(PIN_TEMP3);
97
98     //Status line (always visible)
99 Serial.print("Temperature sensor 1: ");
100 Serial.print(temp1);
101
102 Serial.print("Temperature sensor 2: ");
103 Serial.print(temp2);
104
105 Serial.print("Temperature sensor 3: ");
106 Serial.print(temp3);
107
108
109 Serial.print("Alarm: "); Serial.print(alarmActive);
110 Serial.print("People: "); Serial.println(peopleCount);
111
112 Serial.print(alarmActive ? "ON" : "OFF");
113 Serial.print("People inside: ");
114 Serial.println(peopleCount);
115
116 // Button Logic
117 if (buttonState == LOW && lastButtonState == HIGH){
118     buttonPressTime = millis();
119
120 }
121
122 //if (buttonState == LOW && lastButtonState == HIGH) {
123 if (buttonState == HIGH && lastButtonState == LOW){
124     unsigned long holdfor = millis() - buttonPressTime;
125     if (holdfor >= HOLD_THRESHOLD_MS) {
126         Serial.println(">>> TEST MODE TRIGGERED <<<");
127         runTestMode();
128     } else {
129         if (alarmActive) {
130             Serial.println("MANUAL: Alarm ON");
131             manualAlarm = true;
132             triggerAlarm();
133         }
134         }
135     }
136 }

```

Text

Code

Start Simulation

Send To

1 (Arduino Uno R3)

Text

```

132     if (!alarmActive) {
133         Serial.println("MANUAL: Alarm ON");
134         manualAlarm = true;
135         triggerAlarm();
136     } else {
137         Serial.println("MANUAL: Alarm OFF");
138         manualAlarm = false;
139         clearAlarm();
140     }
141     delay(1000);
142     clearAlarm();
143 }
144 }
145 }
146 }
147 }
148 lastButtonState = buttonState;
149
150 // Sensor Logic
151
152
153 /* here was the primary system which would be more accurate to the product, but sacrificed for the clarity of t
154 bool sensorTriggered =
155     (temp1 >= TEMP_THRESHOLD ||
156     temp2 >= TEMP_THRESHOLD ||
157     temp3 >= TEMP_THRESHOLD);
158
159 if (sensorTriggered && !alarmActive) {
160     Serial.println("!!! FIRE DETECTED - Alarm ON !!!");
161     manualAlarm = true;
162     triggerAlarm();
163 }
164 */
165
166 bool sensor1 = (temp1 >= TEMP_THRESHOLD);
167 bool sensor2 = (temp2 >= TEMP_THRESHOLD);
168 bool sensor3 = (temp3 >= TEMP_THRESHOLD);
169
170 bool sensorTriggered = sensor1 || sensor2 || sensor3;
171
172 if (sensorTriggered && !alarmActive) {
173     Serial.println("FIRE DETECTED Alarm ON !!!!");
174     manualAlarm = true;
175     triggerAlarm();
176 }
177
178 // remember which sensors caused alarm
179 if (!alarmActive) {
180     sensorActive = sensor1;
181     sensorActive = sensor2;
182     sensorActive = sensor3;
183 }
184
185 // Auto-clear only if sensor-triggered (not manual)
186 if (!sensorTriggered && alarmActive && !manualAlarm) {
187     Serial.println("Sensors Clear - Alarm OFF");
188     clearAlarm();
189 }
190
191 // flashing
192 if (!alarmActive) {
193     runFlash();
194 }
195
196 // People Counter
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218 // Other functions to support main code
219
220 float readTemp(int pin) {
221     int raw = analogRead(pin);
222     float voltage = raw * (5000.0 / 1024.0); // in mV
223     return (voltage - 500.0) / 10.0; // TMP36 formula - °C
224 }
225
226 void triggerAlarm() {
227     digitalWrite(PIN_SECONDARY, HIGH);
228     alarmActive = true;
229     peopleCount = 0;
230     Serial.println("People counter reset to 0.");
231 }
232
233 void clearAlarm() {
234     digitalWrite(PIN_SECONDARY, LOW);
235     alarmActive = false;
236     manualAlarm = false;
237
238     digitalWrite(PIN_BUZZ, LOW);
239     digitalWrite(PIN_LEDBLUE, LOW);
240     digitalWrite(PIN_LEDBLUE, LOW);
241     digitalWrite(PIN_LEDGREEN, LOW);
242
243     flashState = false;
244     Serial.print("ALARM CLEARED - People counted: ");
245     Serial.println(peopleCount);
246     peopleCount = 0;
247
248     sensorActive = false;
249     sensorActive = false;
250     sensorActive = false;
251 }
252
253 void runFlash() {
254     if (millis() - lastFlashTime >= FLASH_INTERVAL_MS) {
255         lastFlashTime = millis();
256         flashState = !flashState;
257         digitalWrite(PIN_BUZZ, flashState);
258     }
259
260     // SENSOR 1 - red fast flash
261
262
263
264

```

```

263 // SENSOR 1 - red fast flash
264 if (sensorActive) {
265     digitalWrite(PIN_LEDRD, FLASHSTATE);
266     digitalWrite(PIN_LEDRD, LOW);
267     digitalWrite(PIN_LEDBLUE, LOW);
268 }
269
270 // SENSOR 2 blue flash
271 else if (sensorActive) {
272     digitalWrite(PIN_LEDBLUE, FLASHSTATE);
273     digitalWrite(PIN_LEDRD, LOW);
274     digitalWrite(PIN_LEDBLUE, LOW);
275 }
276
277 // SENSOR 3 - green flash
278 else if (sensorActive) {
279     digitalWrite(PIN_LEDGREEN, FLASHSTATE);
280     digitalWrite(PIN_LEDRD, LOW);
281     digitalWrite(PIN_LEDBLUE, LOW);
282 }
283
284 // default
285 else {
286     if (peopleCount == 0) {
287         digitalWrite(PIN_LEDGREEN, FLASHSTATE);
288         digitalWrite(PIN_LEDRD, LOW);
289         digitalWrite(PIN_LEDBLUE, LOW);
290     }
291     // default
292     else {
293         if (peopleCount == 0) {
294             digitalWrite(PIN_LEDGREEN, FLASHSTATE);
295             digitalWrite(PIN_LEDRD, LOW);
296             digitalWrite(PIN_LEDBLUE, LOW);
297         }
298         else {
299             digitalWrite(PIN_LEDRD, HIGH);
300             digitalWrite(PIN_LEDRD, LOW);
301             digitalWrite(PIN_LEDRD, FLASHSTATE);
302             digitalWrite(PIN_LEDRD, LOW);
303         }
304         digitalWrite(PIN_LEDBLUE, LOW);
305     }
306 }
307
308 void runTestMode() {
309     unsigned long start = millis();
310     while (millis() - start < TEST_DURATION_MS) {
311         digitalWrite(PIN_BUZZ, HIGH);
312         delay(FLASH_INTERVAL_MS);
313         digitalWrite(PIN_BUZZ, LOW);
314         delay(FLASH_INTERVAL_MS);
315     }
316     Serial.println("Test mode complete");
317 }
318
319
320
321
322
323
324
325
326
327
328
329
330

```

*Where the yellow text cut off on page3 of the code ‘sacrificed for the clarity of the simulation’

Concept Evaluation & Summary

The final Concept produced by Pufferfish Security successfully meets the original aim of creating an integrated fire safety system capable of combining multiple fire detection systems and an occupancy sensor. Throughout the project, the team focused on creating a product that would improve emergency response within commercial/industrial buildings by reducing the number of separate systems. By integrating all these sensors, the proposed design offers an efficient and practical solution.

Overall, the project focused heavily on safety, sustainability and practicality. With Low power components, rechargeable batteries and recyclable materials, the company has tried it best to reduce energy consumption and environmental impact. Ethical and legal considerations were all thoroughly addressed maintain privacy and safeguarding standards, ensuring our is clear and customer oriented.

Although there are limitations, particularly the manufacturing cost and the complexity of integrating so many systems into one compact device, the concept remains realistic and feasible since there is no invention of any technology and most exist commercially already. This model was also restricted due to the programs available to us such as tinkercad where we were forced to simplify and imitate different components in the circuit. Further development would require prototyping, testing under extreme environments, code & programming optimisation and aiming to further reduce cost.

In conclusion, the project successfully produces a realistic concept that improves fire detection and public safety. The final designs demonstrate how existing technologies can be integrated into one smart system capable of meeting modern safety requirements.

Appendix

Meeting Notes

9/02/2026

MEETING NOTES BY MARK

5/5 present

Quiz done on suitable roles within a group. Discussed potential roles and responsibilities. Went through the 17 Goals and generated ideas for all of them. Tasks allocated:

- Research into whatever idea that appeals most to you.

Tasks for next week:

- Choosing idea and starting paperwork.

16/02/2026

MEETING NOTES BY MARK

5/5 present

Conferred and collectively decided to choose the idea of a fire alarm with multiple sensors – an all-in-one fire alarm. Expanding this to also include a people counting system. Plan to visit fire station potentially film there, ask firemen for issues and see viability of ideas. Chose to develop on existing idea, not create new item. Motion sensors linked to smoke alarms to detect if people still in the rooms

- Conscious and unconscious
- Separate people from pets and other moving things > potentially pattern recognition.
- May be in addition to CO₂ sensors and other detectors to validate people counter

Decided to do fire safety, mitigation, and prevention Started on to research fire safety and motion sensor idea.

Tasks allocated:

- Research potential technology that could be used
- Research potential risks and problems

Tasks for next week:

- Find solutions to the problems
- Begin official tasks

23/02/2026

5/5 present

MEETING NOTES BY VITALIJA

Found problems with the sensors as it could be considered surveillance, solution was to redirect the audience to public spaces where cameras are already implemented and having the sensor be a relay circuit only turning on when the alarms are triggered. Mollie assigned to tackle safeguarding concerns.

Generated a flow diagram of the circuit by Vitalija. Concluded to have the alarm have a visual and audio output as when power is down, the Bluetooth/Wi-Fi transmission may be unreliable. Discussed Issues with the power economy of the alarm, and potential ways to overcome such as turning the sensor off post sweeps.

Discussed Technology for the sensors and requirements.

Allocated tasks, 1 completer and 1 checker:

Risk assessment	Mark	C. Vitalija
Gant Chart	David	C. Mollie
Safeguarding	Mollie	C. Mark
Work breakdown Structure	Karol	C. David
Market research	Vitalija	C. Karol

Deadline for this on the 23rd of March. Draft due next week. Tasks for next week: consider the 3-minute video and potential ideas.

02/03/2026

4/5 present

MEETING NOTES BY MARK

Completed requirement list Checking people's tasks / completed drafts

Discussed ideas for the video

- 2001 a space odyssey docking scene inspired intro
- Over the shoulder tony stark Esque plans on table
- Joker explosion scene
- Fireman costume

Made basic function- box diagram for alarm system

09/03/2026

3/5 present

MEETING NOTES BY VITALIJA

Compared progress on the tasks discussed deficiencies and next steps as we near the 23rd of March deadline. Dawid also allocated to design an estimated specification for the product. We discussed how we would all prefer a sensor over a camera; however this presents the largest issue.

Sensors vs cameras:

- People vs hot items etc > ai
- Safeguarding
- Cost > Aim for under £100 pounds
- Range
- Connectivity – large warehouses

We have decided to try and produce a sensor that complies with all these requirements. If this proves to fail, we will resign to use a camera.

Tasks:

- Using the market research, find similar products and how they work
- Find how we can design an efficient people counter / sensor without a camera

16/03/2026

4/5 present

MEETING NOTES BY MOLLIE

Started work on the morphological chart

Research on components of the morphological chart including:

- Types of gas detectors
- Types of heat detectors
- Types of smoke detectors
- Types of motion sensor
- Types of people counters
- Types of speakers
- Types of lights

Started the function means tree.

23/03/2026

5/5 present

MEETING NOTES BY VITALIJA

Found a very useful document containing all the studies we need, the main problem is the cost. Aim today is to try and find a way to reduce this number as much as we can.

Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios - PMC

Due next week Monday:

- Finished docs
- Consider sketch/ physical format/casing
- Perks & rando stuff
- Technology
- Sensor selection& coding
- Cost consideration

20/04/2026

5/5 present

MEETING NOTES BY VITALIJA

Finished the allocated tasks of researching technology and construction of the alarm system. We discussed every possible variation of the alarm using the system tree and discussed each pro and con. We discussed what we must consider for the alarm including power economy and aesthetics. Next task allocated is to each design a realistic draft of the alarm.

27/04/2026

5/5 present

MEETING NOTES BY VITALIJA

Having designed a draft each, we collated our plans into one simple plan. We have decided to use a plastic fire-resistant casing, a photoelectric smoke sensor to alert slow rising fires, MOS detector for CO and MO detection as it is most power efficient, low wave infrared as it proves to be the most efficient financially and is reliable in fires and hot environments. We save money by using a reliable but cheap heat sensors, and a simple buzzer and LED. These sensors must all be up to code and regulation. We are now starting to put together the portfolio and assigning sections to each person.

4/05/2026

5/5 present

MEETING NOTES BY VITALIJA

Having put together most of the portfolio, we are looking to ensure its organised in a logical structure. We have brainstormed ideas for the video and plan to use the university facilities to keep the video engaging and demonstrating the kind of buildings that the alarms could be used in. To do: Complete AutoCAD & Circuit Models – Dawid & Vitalija, Complete Material List – Mollie, Begin & Edit video – Mark, Complete WBS and Team Role & Dynamics for Portfolio – Karol, & Submit work in- vitalija.

References:

Research paper on thermal imaging

Tsai, P.-F., Liao, C.-H. and Yuan, S.-M. (2022) 'Using deep learning with thermal imaging for human detection in heavy smoke scenarios', *Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios.*, 22(14), p. 1. doi:10.3390/s22145351.

Market Research:

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Battery heat detector – 10-year battery & RF interconnectivity (2025) Hispec. Available at: <https://hispec.co.uk/product/product-radio-frequency-lithium-battery-product-radio-frequency-lithium-battery-heat-detector-powered-by-a-10yr-sealed-lithium-battery-smoke-detector-powered-by-a-10yr-sealed-lithium-battery/> (Accessed: 13 May 2026).

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